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# RV COLLEGE OF ENGINEERING®

**(An Autonomous Institution Affiliated to VTU)**

**VI Semester B. E. Regular Examinations August-2025**

**Computer Science and Engineering**

## NETWORK PROGRAMMING AND SECURITY

***Time: 03 Hours*** ***Maximum Marks: 100***

***Instructions to candidates:***

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

### PART-A

| | | | M | BT | CO |
|---|---|---|---|---|---|
| 1 | 1.1 | Justify the purpose of four-way handshake of SCTP compared to three-way of TCP. | 02 | 2 | 1 |
| | 1.2 | Can the server rebind to port 8080 immediately? What is the condition? | 02 | 1 | 1 |
| | 1.3 | Name the functions that pass a socket address structure from the kernel to the process. | 02 | 1 | 2 |
| | 1.4 | A concurrent TCP server uses fork() to handle clients. After the server accepts a connection and forks, which sockets(s) must be closed in the child and parent processes? | 02 | 1 | 2 |
| | 1.5 | Mention the importance of gai_strerror with example. | 02 | 2 | 3 |
| | 1.6 | How does a UDP client/server deal with the scenario's of lost datagram at client/server end? Identify one way to handle this situation. | 02 | 3 | 3 |
| | 1.7 | Mention any two factors that increases computation time in Fiestel Cipher. | 02 | 2 | 4 |
| | 1.8 | What is the block size in DES? What is the cipher key size in DES? What is the round-key size in DES? | 02 | 1 | 4 |
| | 1.9 | List any two of the four security protocols at transport layer by identifying the purpose of these protocols as defined in SSL or TLS. | 02 | 1 | 4 |
| | 1.10 | Identify by listing any two security threats to wireless networks. | 02 | 3 | 4 |

### PART-B

| | | | | | |
|---|---|---|---|---|---|
| 2 | a | With a suitable timeline diagram explain how SCTP association is established? | 05 | 2 | 1 |
| | b | Discuss port numbers used by the transport layer protocols differentiate its types highlighting the significant aspects of each type. | 06 | 3 | 1 |
| | C | Why is the TIME_WAIT in TCP typically set to 2 RTT (Round-Trip Time)? What would happen if the timeout was set to less than 2 RTT or more than 2 RTT? | 05 | 3 | 1 |
| | | | | | |
| 3 | a | Describe the approach of concurrent server for handling multiple clients. | 08 | 2 | 1 |
| | b | In a TCP client-server application, both getsockname() and getpeername() functions are called on a connected socket (connectfd) in the child process. Explain what each function returns and under what circumstances the two might return the same IP address and port number. Justify your answer with a brief explanation. | 08 | 5 | 2 |

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|---|---|---|---|---|---|
| | | **OR** | | | |
| 4 | a | Discuss with a diagram, describe a simple echo client and server. Write C program for TCP echo server. | 10 | 3 | 2 |
| | b | Demonstrate the working of the EXEC system call by creating child process that executes a different program, passing arguments to the main() function of the new program. | 06 | 3 | 2 |
| | | | | | |
| 5 | a | With a neat diagram explain DNS and Resource Records. | 10 | 2 | 3 |
| | b | Illustrate DNS function which can resolve both IPv4 and IPv6 with syntax. Show how to pass TCP and IPv4 in hints structure. | 06 | 2 | 3 |
| | | **OR** | | | |
| 6 | a | Write UDP client and server program where client reads sends the Audio file name to the server and server reads the content of the file name sent by client and sends the content of the Audio file to client. | 08 | 2 | 3 |
| | b | Illustrate with neat figure UDP client and server by identifying the socket API's required at the client and server respectively, | 08 | 2 | 3 |
| | | | | | |
| 7 | a | Differentiate between Stream and Block Ciphers. Include examples or scenarios where stream ciphers are preferred over block ciphers and vice versa. | 06 | 2 | 4 |
| | b | Explain with neat diagram DES construction in detail with a neat diagram. | 10 | 2 | 4 |
| | | **OR** | | | |
| 8 | a | Explain in detail Feistel Block Cipher structure with neat sketch. | 08 | 2 | 4 |
| | b | Describe RSA algorithm and Estimate the encryption and decryption values for the RSA algorithm parameters. | 08 | 2 | 4 |
| | | | | | |
| 9 | a | With a neat timeline diagram illustrating the initial exchange needed to establish a logical connection between client and server, discuss briefly the Handshake Protocol that SSL defines to authenticate the server and the client. | 10 | 2 | 4 |
| | b | With a neat structure for IEEE 802 protocol architecture, explain IEEE 802 standards within the structure of a layered set of protocols. | 06 | 2 | 4 |
| | | **OR** | | | |
| 10 | a | With an appropriate illustration, discuss the overall operation of SSL Record Protocol. | 08 | 3 | 4 |
| | b | Describe the phases of operation of IEEE 802.11i with a neat diagram representing the different phases of operation. | 08 | 2 | 4 |